KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4115
Computer and Network Security

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "Asymmetric cryptosystems are based on clever application of number theory concepts" – individually establish it for RSA, ElGamal and Paillier cryptosystems about different keys. (12) [CO2]
   b) Establish usages of both block and stream ciphers in cases of two asymmetric and one symmetric cryptosystems. If not possible to deploy, also establish it. (12) [CO3]
   c) Discuss properties of digital signatures. Analyze them about ElGamal digital signature. (11) [CO2]

2. a) Discuss characteristics of cryptographic hash functions. Digital signature, asymmetric cryptosystem and hash function need to combine concurrently – why and how? (13) [CO1]
   b) Define and discuss characteristics of 'Zero Knowledge Proof'. Using a block diagram depict the symmetric cryptosystem along with the ways of the key distribution. (14) [CO2]
   c) Derive a bonding among order, prime, and primitive element considering the prime 13 and order 2. (08) [CO4]

3. a) Explain how triple DES performs data encryption. Using a block diagram illustrate the major steps of each cycle of AES algorithm. (11) [CO1]
   b) Discuss the following attacks: (i) Ciphertext-only, (ii) Chosen-plaintext, (iii) DoS, and (iv) DDoS. How Known-plaintext-attack can be mounted on RSA cipher? Using the required parameters, derive the mechanism. (15) [CO4]
   c) Define Firewalls. Discuss major types of Firewalls. Discuss various network threats elaborately. (09) [CO2]

4. a) Describing required conditions and parameters, discuss how RSA and ElGamal cipher can be used to develop product cipher. Derive re-randomization property of ElGamal cipher. (15) [CO3]
   b) 'Confusion' and 'Diffusion' are the ultimate goal of every cryptosystem – why? Discuss Shannon characteristics of good cipher. (09) [CO1]
   c) Establish a comparison between AES and DES. Show that the encryption and decryption operations of Caesar cipher work according to substitution cipher. (11) [CO2]

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define the Discrete Logarithm Problem (DLP) and explain why it is considered computationally difficult to break in cryptography. (09) [CO1]
   b) In the Diffie-Hellman Key Exchange (DHKE) protocol, each participant selects a private key 'a' from the range $\{2, \dots, P\text{-}2\}$ where P is a prime number. Explain why the values $a = 1$ and $a = P - 1$ are excluded from this range. (11) [CO2]
   c) Explain scalar multiplication operation on Elliptic Curve (EC). Given an EC (15) [CO3]
   $$E: y^2 = x^3 + 3x + 5 \bmod 17$$ and point $P = (1,3), Q = (4,8)$,
   compute $2P = P + P = (x_3, y_3)$ and $P + Q = (x_4, y_4)$. Consider the equations:
   $$x_3 = S^2 - x_1 - x_2 \bmod P$$
   $$y_3 = S(x_1 - x_3) - y_1 \bmod P$$
   $$S = \begin{cases} \dfrac{3x_1^2 + a}{2y_1} \bmod P \\ \dfrac{y_2 - y_1}{x_2 - x_1} \bmod P \end{cases}$$

CamScanner

6. a) Explain Post-Quantum Cryptography (PQC) and discuss its impact on symmetric and asymmetric cryptographic algorithms. (07)
   b) Explain the DHKE protocol and its key setup. Describe the EC-DHKE protocol and compare it with standard DHKE in terms of security and key size. (13) [C
   c) Define Transport Layer Security (TLS) and its purpose. Describe the TLS 1.2 handshake process. Compare TLS 1.2 and TLS 1.3 in terms of handshake efficiency and latency. Is O-RTT handshake supported in TLS 1.3, and under what conditions? (15) [CO4

7. a) Explain the N^2 key distribution problem in symmetric key systems. (05) [CO1]
   b) Define Elliptic Curve Discrete Logarithm Problem (ECDLP) and explain why computing it is considered difficult. (05) [CO2]
   c) Explain Elliptic Curve Cryptography (ECC) by describing its key generation, encryption and decryption processes. (10) [CO3]
   d) Illustrate the following web attacks with diagrams showing attacker-victim-server interactions, and explain effective mitigation techniques for each: (15) [CO4]
      i) Cross-site Request Forgery (CSRF).
      ii) Server-side Request Forgery (SSRF).
      iii) Cross-site Scripting (XSS).

8. a) Various malicious codes have different characteristics – discuss them briefly. Explain characteristics of good viruses. (05) [CO2]
   b) Define the Point at Infinity in ECC and explain the conditions under which it is reached. (06) [CO2]
   c) Explain homomorphic encryption and its types. Derive the equations of homomorphic operations in the Paillier cryptosystem and illustrate each with an example. (12) [CO3]
   d) Illustrate a Man-in-the-Middle (MITM) attack on the DHKE protocol and discuss methods to prevent such attacks in public key cryptosystems. (12) [CO3]



CamScanner